

Real-World Application: Online Video Streaming

1. Parallel Computing

How it's used: In the realm of online streaming, parallel computing plays a crucial role in video encoding and processing. Platforms like Netflix and YouTube employ parallel computing techniques to compress, transcode and ready video files for streaming in various resolutions - ranging from 480k to 4K. They break down large video files into smaller segments allowing multiple CPUs or GPUs to work on them at the same time, which greatly accelerates the encoding process.

Why it's important

Without the efficiency of parallel processing, encoding each video would take an impractical amount of time, particularly given the vast number of videos available. This technology ensures that content can be accessed quickly and in multiple formats, tailored for different devices and bandwidth scenarios.

2. Networking Systems

How it's used: Streaming services depend on extensive networked systems that include data centers and content delivery networks (CDNs). When a video is streamed, it's usually pulled from the nearest server to the user, which helps to minimize latency and enhances streaming quality. These interconnected systems work in harmony to efficiently manage traffic and balance loads.

Why it's important: Networked systems are essential for delivering real-time video to audiences around the globe. They guarantee high availability, and minimal buffering by intelligently distributing traffic and data to optimize user experience.